

MoE Model Exit Exam-Chemical Engineering

1. Priming in centrifugal pumps is_____
 - a) Cleaning the pump before starting the pump
 - b) Removing the liquid in the suction side of the pump
 - c) Checking the appropriateness of the operation
 - d) Filling the suction side of the pump with liquid
2. Which one of the following is not true about ideal solutions
 - a) No absorption or release of heat occurs during mixing
 - b) They follow Raoult's law
 - c) Excess properties are zero for ideal solutions
 - d) The entropy change of mixing is zero
3. Which dimensionless number is commonly used for high-speed compressible flow
 - a) Reynolds number
 - b) Mach number
 - c) Euler number
 - d) Froude number
4. Find the dumping factor (ζ) for a process whose transfer function is
$$G(s) = \frac{K}{(s+1)(2s+1)}$$
 - a) $3/\sqrt{2}$.
 - b) $3/(2/\sqrt{2})$
 - c) $2/(3/\sqrt{2})$
 - d) 4
5. Codes and standards used in engineering practice do not include one of the following.
Which one?
 - a) Testing procedures for performance, compositions, quality.
 - b) Operating manuals
 - c) Materials and their properties
 - d) Preferred sizes for tubes, plates, etc.
6. Which of the following is wrong?
 - a) Multiple phases at the same T and P are in equilibrium when the chemical potential of each species is the same in all phases
 - b) Coexisting liquid and vapour phases of a pure species are said to be in equilibrium when they have the same temperature, pressure, and fugacity
 - c) For coexisting vapour and liquid phases of a pure species their fugacity coefficients should not be equal.
 - d) In coexisting liquid and vapour phases of a pure species, there is movement of molecules from one phase to other

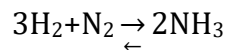
7. Which of the following relations is wrong?
- a) Reynolds number- flow regimes in incompressible
 - b) Weber number-liquid atomization
 - c) Froude number- heat transfer
 - d) Mach number- compressibility effects in a fluid flow
8. Which type of centrifugal pump delivers higher specific speed
- a) Axial
 - b) Radial
 - c) None
 - d) Mixed
9. Which of the following is correct for an exothermic reaction?
- a) The reactants have less energy as compared to the products
 - b) The reactants have more energy as compared to the products
 - c) Both reactants and products have same energy
 - d) The products have more number of moles as compared to the reactants
10. Consider a double pipe heat exchanger in which the hot liquid flows through the outer pipe and the cold fluid through the inner pipe. Which of the following is wrong for the case under consideration?
- a) Overall heat transfer coefficients depend on which area we select for the h at transfer equation
 - b) The heat transfer rate can be obtained from the ratio of the driving force to the resistance.
 - c) The driving force is obtained from the logarithmic mean temperature difference.
 - d) A single overall heat transfer coefficient that is independent of heat transfer area can be obtained.
11. Which of the following least favours the selection of a batch process over a continuous process?
- a) If the process is expected to cause severe fouling that require frequent cleaning
 - b) If the product is single
 - c) If the catalyst you use is short life
 - d) If your project has design uncertainties
12. Which of the following is not true about urea production?
- a) Cannot be a raw material for urea synthesis
 - b) NH_2CONH_2 is the chemical formula of urea
 - c) Natural gas can be a raw material for urea synthesis
 - d) Urea is formed from the reaction between CO_2 and NH_3
13. Which of the following is not correct?
- a) Cavitation is more likely to occur with hot liquids.
 - b) Cavitation is more likely to occur with high-speed pumps.

- c) Cavitation is more likely to occur on the discharge side of centrifugal pump compared to the suction side
 - d) Cavitation is s more likely to occur when the pressure in the liquid is less than the vapour at the liquid temperature
- 14.** Which one of the following is true about constant rate filtration?
- a) Permeate flow rate progressively decreases
 - b) They are the most common filtration operations
 - c) Pressure drop is held constant
 - d) Pressure drop progressively increases
- 15.** Which one of the following is a process with a time delay?
- a) $K/((s+1))$
 - b) K/s
 - c) $K/(s+1) (2s+1)$
 - d) $(K[(e)]^{(-\theta s)})/((S+1))$
- 16.** What is the role of the catalysts in a chemical reaction?
- a) to involve in the chemical reaction as part of reacting species
 - b) to serve as a media for the reaction
 - c) to increase the equilibrium conversion of a reaction
 - d) provide an alternative reaction path with lower activation energy
- 17.** Which one of the following does not affect the duration of the lag phase?
- a) The number of inoculated cells
 - b) The age of inoculated cells
 - c) The type of cells
 - d) None
- 18.** What is the use of a transducer in process control?
- a) To measure pressure
 - b) To compare measured value of the output with the set point
 - c) To convert a signal from one form to another, for e.g., from current to pressure
 - d) To measure temperature
- 19.** Which one of the following is true about friction factor for fully turbulent flow in a pipe
- a) it is dependent only on the relative roughness
 - b) it is dependent on viscosity only
 - c) it is dependent on Reynolds number and relative roughness
 - d) None
- 20.** Derivative control is not advised for controlling_____.
- a) Flow rate
 - b) Temperature control in an endothermic reactor
 - c) Level control in a large tank with big diameter
 - d) Temperature variations in a jacketed heater

21. The pressure exerted by an incompressible static fluid on its container is dependent on
- the depth of the fluid in the container
 - the base area of the container
 - temperature of the fluid in the container
 - the volume of the container
22. Which one of the following is the most appropriate for measuring very small pressure difference?
- Inclined tube mercury
 - Venturi meter
 - U-tube water
 - Inclined tube water.
23. Which of the following is not a correct representation?
- For Proportional Integral-Derivative controller: $G_C(s) = K_c (1 + 1/(T_I S) + T_D S)$
 - For Proportional-Derivative controller: $G_C(s) = K_c (T_D S)$
 - For Proportional controller: $G_C(s) = K_c$
 - For Proportional Integral- controller: $G_C(s) = K_c (1 + 1/(T_I S))$
24. Which one of the following is mostly observed way of replenishing the dissolved oxygen in water stream that is polluted with industrial waste?
- By adding oxygen using chemicals
 - Eutrophication
 - Natural aeration of water stream from the atmosphere.
 - Photosynthetic action of algae.
25. How is the size of the internal diameter of a distillation column estimated?
- Based on the superficial vapour velocity
 - Based on the difference between outside and inside
 - Based on the reflux ratio
 - Based on the selected plate/tray type
26. At high Reynolds number
- Forces are unimportant and viscous forces control.
 - Inertial forces control and viscous forces are unimportant.
 - Viscous forces predominate.
 - None
27. In the radial flow pump characteristic curves shown below, which point indicates the optimum point for operation (duty point)
- Point D
 - Point A
 - Point C
 - Point B

28. Which one of the following is most likely practiced during sterilization by using heat?
- The number of contaminant cells must be reduced to absolute zero.
 - The number of contaminant cells must be reduced by five order of magnitude.
 - The number of contaminant cells must be reduced to a very low number that is considered safe.
 - The number of contaminant cells must be reduced by two order of magnitude
29. Which one of the following is appropriate rate expression for the conversion of reactant A in a batch reactor? Where V -reactor volume, N_A -number of moles of A, N_{Ao} - input molar flow rate of A, N_A -molar flow rate of A leaving the reactor, S -cross sectional area of the reactor perpendicular to the flow direction, x -axial length of the reactor
- $1/S[(dN)]_A/dx$
 - $(N_{Ao}-N_A)/V$
 - $1/V(dN_A)/dt$
 - None
30. What do you think happens to the temperature of an air inside a completely closed and insulated room if a heater which is plugged in the inside wall of the room is turned on?
- Room air temperature raises due to an energy generation inside the room
 - Room air temperature raises due to an energy input to the system in the form of heat
 - Room air temperature will not be affected as there is no energy crossing the boundary
 - Room air temperature raises due to an energy input to the system in the form of work
31. Consider the airlift bio reactor shown in the Figure. Why does the liquid in the down corner flow downwards and the liquid in the central section upwards?
- because temperature differences of the liquid-gas mixtures in the two sections
 - because of bulk density differences of the liquid-gas mixtures in the two sections
 - because of the presence of a circulation pump
 - None of the above
32. Based on the method of McCabe-Thiele, how is the feed point identified?
- From the intersection of the q -line and the upper operating line
 - From the intersection of the q -line and the two operating lines
 - From the intersection of the upper and lower operating lines
 - From the intersection of the q -line and the lower operating line
33. Which one of the following is not true statement?
- Raoult's law works well in all temperature range
 - Summability relations are used to calculate total solution properties from partial properties
 - Raoult's law considers ideal gas in the vapor phase and ideal solution in tile liquid phase.

- d) Partial properties are mixture properties derived from total solution properties
- 34.** Identify the correct statement
- a) Generally, as pressure increases, the boiling point of a substance decreases
 - b) As pressure is increased the melting point of a substance increases but the boiling point decreases
 - c) Generally, as pressure increases, the melting point of a substance also increases
 - d) As pressure is increased the melting point of a substance decreases
- 35.** Fluid that has higher tendency to foul heat transfer is placed in the tube side flow. Why?
- a) It is easier to clean tubes
 - b) Replacing tubes is technically and economically
 - c) Allowable flow rate is higher in the tube which can reduce fouling
 - d) All
- 36.** Which one of the following doesn't favor the forward reaction shown below?



- a) Reducing the amount of nitrogen
 - b) Introducing an inert gas to the system
 - c) Increasing pressure
 - d) Continuously removing the ammonia
- 37.** Which of the following does not need sterilization before the start of fermentation?
- a) the culture medium
 - b) the equipment used for fermentation
 - c) air supplied to the fermenter (in case of aerobic fermentation) ,
 - d) None
- 38.** One of the following is not considered as criterion for selecting refrigerant to be used in a heat pump.
- a) Cost
 - b) Toxicity
 - c) Corrosion properties
 - d) The coefficient of performance
- 39.** Suppose cells already growing in the exponential growth phase are inoculated into a medium of the same composition. Then, which one of the following will most likely takes place?
- a) The lag phase will not be affected.
 - b) The lag phase may become shorter.
 - c) The inoculated cells will die immediately.
 - d) The lag phase may become longer.
- 40.** Which one of the following is not true about controller effect on the controlled process?
- a) controller eliminate offset
 - b) Proportional-only controller eliminate offset

- c) Proportional-only controller speed up the response
 - d) Proportional-integral controller increase the overall order become sluggish (slow) the overall response
- 41.** Considering all other conditions are equal, which of the following is correct regarding co-current and counter-current heat exchanger arrangement?
- a) There is a possibility for the outlet temperature of the cooling liquid to be higher than the outlet temperature of the heating liquid in counter-current flow heat transfer as compare to co-current heat transfer arrangement
 - b) The log-mean temperature difference of co-current flow is higher than that of the counter-current flow for the same inlet and outlet conditions of the two fluids.
 - c) Co-current flow heat transfer arrangement allows extraction of a higher proportion of the heat content of the heating fluid as compared to counter-current flow
 - d) The required heat transfer area in counter-current flow is higher than that in co-current flow for the same inlet and outlet conditions of the two fluids.
- 42.** Which of the following is true about Prandtl number (Pr)
- a) it is the ratio of thermal diffusivity to mass diffusivity
 - b) it is the ratio of conductive flow to viscous flow
 - c) it is the ratio of inertial force to viscous force
 - d) it is the ratio of momentum diffusivity to thermal diffusivity
- 43.** Identify the incorrect statement about packed bed reactors
- a) They are good reactor systems for heterogeneous catalytic reaction
 - b) They give maximum conversion per weight of catalyst
 - c) The packing are usually solid catalysts
 - d) They are good for temperature control
- 44.** Identify the true statement about net positive section head (NPSH) of a pump
- a) A large slow moving pump requires higher NSPH than a small fast moving pump
 - b) It is independent of discharge pressure
 - c) It is independent of the impeller speed
 - d) It is the amount by which the pressure at the section point of the pump, expressed as a head of the liquid to the pump must exceed the vapor pressure of the liquid
- 45.** Which of the following is correct?
- a) For exothermic reaction to be carried out isothermally, heat must be supplied to the system.
 - b) If an endothermic reaction is to be carried out isothermally, heat must be removed from to the system.
 - c) In an endothermic reaction is carried out adiabatically, the product temperature will increase.
 - d) If combustion is to be carried out adiabatically, the product temperature will rise.

46. Consider the system shown below servo problem case. Let, $G_m = G_c = G_f = 1$, and $G_p = \frac{1}{(\tau_1 s + 1)(\tau_2 s + 1)}$

The system is stable,

- a) If τ_1 and τ_2 are both positive
 - b) If τ_1 and τ_2 are both negative
 - c) If τ_1 is positive and τ_2 is negative
 - d) If τ_2 is positive and τ_1 is negative
47. Which one of the following does not describes chemical potential?
- a) It is a criteria for mechanical equilibrium
 - b) It is a partial property
 - c) It is differential change in Gibbs free energy of a system with respect to the differential change in one component of a system keeping other variable constant
 - d) It is a criteria for phase equilibrium
48. Identify the correct statement about residual and excess properties
- a) excess properties are the difference in real gas and ideal gas properties
 - b) None
 - c) residual properties are the difference between properties
 - d) both are the difference in real and ideal system properties
49. The Frasch process is for mining of sulphur as the Solvay process for ____.
- a) producing nitric acid
 - b) manufacture of soda ash
 - c) producing iron from iron ore
 - d) manufacture of ammonia
50. Which one of the following is true?
- a) Latent heat is accompanied by temperature change
 - b) The heat transferred to or from a system during phase transformation is called sensible heat
 - c) Phase transformation is an isothermal process
 - d) Ideal gas specific heat capacities are functions of pressure and temperature
51. Activity coefficient is the measure of
- a) partial volume of liquid
 - b) departure from ideal gas behavior
 - c) vapor pressure of a liquid
 - d) departure from ideal solution behavior
52. Which of the following is not correct?
- a) Environmental concerns constrain the selection of compounds to one used as refrigerants.
 - b) Vapor pressure of a refrigerant should be less than atmospheric pressure
 - c) The higher the vapor pressure of a refrigerant is, the higher will be the cost of the condenser.
 - d) Carnot heat pump is independent of the refrigerant type used in the system.